

StarAnalytics

Design & Product Requirements

Nivin Pauly · Social Media Intelligence Platform

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1. Executive Summary

StarAnalytics is a private web dashboard purpose-built for Nivin Pauly's social media management team. It replaces manual reporting across multiple tools (Instagram Insights, spreadsheets, third-party trackers) with a single, actor-centric workspace that covers page analytics, campaign intelligence, agency performance scoring, and fan community monitoring.

This document defines every screen, feature, data requirement, and technical build path needed to move from the current working HTML prototype to a production-grade deployment.

Core Value Proposition

One dashboard that tells the team: what is performing, which campaign is gaining traction, which agency is delivering genuine reach, and which fan pages need engagement — all without switching between tools.

2. Problem Statement

Current pain points identified:

- Instagram Insights only shows own-page data — no competitor benchmarking, no hashtag campaign view
- Campaign monitoring is manual: team members check hashtags on phone and paste numbers into spreadsheets
- 10 agencies submit reports separately; no single view to compare performance or flag inauthentic engagement
- Fan pages are tracked by memory — no alert system when a fan page posts a critical campaign hashtag
- No sentiment signal: the team doesn't know if a campaign is creating positive or negative chatter

3. Product Scope & Screens

The application is structured into six top-level sections accessible from a persistent sidebar. The Campaigns section expands into three inner tabs.

3.1 Sidebar Navigation

My Page	Dashboard, Content, Audience, Compare Pages
Campaigns	Own Campaigns · Hashtag Search · Agency Report
Community	Fan Pages (24 tracked)

3.2 Dashboard

The landing screen. Displays a four-metric KPI row (Followers, Engagement Rate, 30d Reach, Profile Visits), a 12-week follower growth line chart, an engagement-type doughnut (Likes / Comments / Saves / Shares), and a grid of the top four posts by engagement this month.

Design Note

All chart colours follow Instagram brand palette: #E1306C (primary pink/red), #833AB4 (purple), #F77737 (orange). Charts are rendered via Chart.js 4.x CDN.

3.3 Content

Format-level performance view. Insight row shows best-performing format (Reels at 4.8% avg engagement), best posting time (Friday 7 PM), posts published this month, and average reel views. Two charts: post-type bar (Reels / Photos / Carousels / Stories by engagement %) and weekly reach trend line. Below: 4×2 scrollable post grid with engagement stats on hover.

3.4 Audience

Three KPI chips: top city (Kochi, 31%), dominant age group (25–34, 44%), gender split (64M / 36F). Location section shows horizontal bar chart of top 5 cities. Age distribution is a horizontal bar chart. Below both sits a Best Time to Post heatmap: 5 days × 6 time slots, five heat levels using a pink ramp (#f0f0f5 → #880e4f).

3.5 Compare Pages

Side-by-side metric comparison. Default comparison: Nivin Pauly (7.4M) vs Dulquer Salmaan (9.1M). Metrics: Followers, Engagement Rate, Avg Likes/Post, Posts/Week, Reel Avg Views, Story Response Rate. Win cells highlighted in green. "Add account to compare" button connects to additional pages via the Instagram Graph API in production.

3.6 Campaigns — Own Campaigns

Lists all active and planned campaigns as cards. Each card shows: campaign name, status pill (Live / Planned), date range, hashtags, and summary stats. Clicking a Live campaign opens the campaign detail view.

#vijayam Detail View

Hero section with tag, reach, posts-today, +posts-last-hour, and unique accounts stats. KPI row: avg engagement 8.4%, 1,092 reels with tag, 612 stories with tag, 78% positive sentiment. Below:

- Hourly volume sparkline (since announcement)
- Sentiment bar: 78% positive / 14% neutral / 8% negative
- Geographic spread: Kerala 44%, Tamil Nadu 26%, UAE/Gulf 16%, Karnataka 8%, Others 6%
- Top keyword pills (excited, fire, blockbuster, can't wait, goosebumps, masterpiece)
- Live post stream with fan page badges, handle, timestamp, text, and per-post stats

3.7 Campaigns — Hashtag Search

Search input + Track button. Below: table of tracked hashtags showing name, reach bar, post count, engagement %, and status pill (New / Trending / Campaign / Active). Eight hashtags tracked by default matching active campaigns.

3.8 Campaigns — Agency Report

A four-view flow activated within the Agency Report tab:

- Upload — drag-and-drop zone for Excel/CSV or direct URL paste; 10 agencies listed with link counts (50 each = 500 total); "Analyse All Posts" CTA
- Agency Scorecard — five KPI chips; ranked leaderboard with colour-coded scores; score breakdown card (Performance 50% / Authenticity 30% / Efficiency 20%); campaign health bars
- Authenticity Audit — four flag types with agency attribution; per-agency authenticity bar chart; positive signals panel
- All 500 Posts — sortable, filterable table with URL, agency, reach, likes, comments, saves, engagement %, authenticity score, overall score, and flag chips

3.9 Fan Pages

Tracks 24 registered fan pages. KPI row: 4.8M total fan reach, 18/24 active today, 21/24 posting #vijayam. Filter tabs: All / Most Active / Posting #vijayam / Largest Reach. Each fan row: avatar, name, handle, followers, engagement rate, posts today, mini sparkline, live/idle dot, #vijayam pill. Alerts panel at bottom with bell icons and timestamps.

4. Agency Scoring System

4.1 Scoring Formula

Dimension	Weight
Performance	50%
Authenticity	30%
Efficiency	20%

4.2 Authenticity Flags

Flag Type	Description
Engagement velocity anomaly	Likes spike > 10K within 2 min of posting
Generic comment pattern	Single emoji or copy-paste comments from 0-post accounts
Low save-to-like ratio	High likes but near-zero saves
Off-hours engagement spike	Significant engagement between 2–4 AM IST

4.3 Agency Leaderboard — Current Run

#	Agency	Score	Perf	Auth	Eff	Flags	Status
1	Pixelwave Media	91	94	90	89	0	Clean
2	Reelcraft Agency	86	88	87	83	1	Clean
3	Kochi Digital Co.	82	84	83	79	0	Clean
4	SocialSpark IN	79	80	82	76	1	Clean
5	TrendTide Media	76	77	78	74	0	Clean
6	CreatorHub	74	75	76	71	0	Clean
7	Viralthread MKT	71	73	70	69	2	Review
8	BuzzBridge	58	72	38	64	23	FLAGGED
9	Influx Kerala	52	65	32	60	18	FLAGGED
10	NovaSocial	48	60	28	55	13	FLAGGED

BuzzBridge, Influx Kerala, and NovaSocial are flagged for authenticity issues. Recommend withholding payment pending re-verification or replacing with top-tier agencies.

5. Data Requirements

5.1 Instagram Graph API Endpoints

Endpoint	Data Retrieved
/me/insights	Impressions, reach, profile views, follower count trend
/me/media	Posts, media type, thumbnail, caption, timestamp
/media/{id}/insights	Likes, comments, saves, shares, reach per post
/me/audience_demographics	Age, gender, city, country breakdowns
/ig_hashtag_search	Hashtag ID lookup for campaign tracking
/[{hashtag_id}]/recent_media	Recent posts using hashtag (public only)
/ig_user/{id}	Competitor page follower count, media count

API Rate Limit Note

Instagram Graph API rate limits: 200 calls per user per hour for basic display, 4,800/day for business accounts. Agency post scraping (500 posts) must use a third-party scraper (Apify / CrowdTangle) as the Graph API does not provide access to other users' post-level metrics.

5.2 Third-Party Data Sources

- Apify Instagram Scraper — for hashtag monitoring, fan page scraping, agency post metrics beyond Graph API limits
- Google Natural Language API — for Malayalam/English sentiment analysis on post captions and comments
- Custom post ingestion job — accepts Excel/CSV upload or URL list, queues Apify scrape, stores results in DB

6. How to Build It — Technical Insights

6.1 Recommended Stack

Layer	Technology
Frontend	Next.js 14 (App Router) + TypeScript
Charts	Chart.js 4 via react-chartjs-2
Styling	Tailwind CSS + CSS variables
Backend	Node.js / Fastify API
Database	PostgreSQL (via Supabase)
Auth	NextAuth.js + Google OAuth
Scraping	Apify Cloud (IG scraper actors)
Sentiment	Google NL API
Hosting	Vercel (frontend) + Railway (API)
File parsing	xlsx npm package

6.2 Build Sequence (Recommended)

Follow this order to deliver usable value at each step while minimising rework:

1. Auth + shell — Next.js project, Tailwind, sidebar nav, screen routing. Mirrors prototype HTML structure directly.
2. Dashboard screen with real IG Graph API data. Connect /me/insights and /me/media. Wire Chart.js growth + engagement charts.
3. Content + Audience screens. Both pull from the same Graph API endpoints — aggregate by media_type for Content, demographics for Audience.
4. Compare Pages. Add account lookup via Graph API; store competitor IDs in DB.
5. Campaigns CRUD — database schema, Own Campaigns list, campaign detail. Hashtag polling via Apify; store raw posts in Postgres.

6. #vijayam live view — real-time updates via Supabase subscriptions or polling. Sentiment via Google NL API per post.
7. Agency Report — file upload endpoint, Apify batch job, scoring engine, results stored per campaign run.
8. Fan Pages — manual add + auto-detect from hashtag stream, sparkline from last 7 days of post data, alerts via email/push.

6.3 Key Technical Decisions

Real-time vs Polling

True real-time (WebSocket) is overkill for MVP. Use a 15-minute polling cron job for hashtag volume. Reserve WebSocket for the live post stream on the campaign detail page once data volume justifies it.

Agency Scraping Architecture

On "Analyse All Posts" click: frontend sends 500 URLs to API → API enqueues Apify actor run → returns run ID → frontend polls /api/agency-run/{id}/status every 5s → on completion, results written to DB and scorecard rendered. Show a progress indicator during analysis.

Authenticity Engine

Implement as a pure function: `scorePost(post) → { performanceScore, authenticityScore, efficiencyScore, flags[] }`. Keep it deterministic and unit-testable. Version-control the thresholds as config so the team can tune them without code changes.

Malayalam Sentiment

Google NL API supports Malayalam natively since 2022. Pass caption + top 20 comments per post. Cache results in DB by post ID — comments don't change significantly after 24 hours, so re-scoring is wasteful.

6.4 Database Schema (Core Tables)

Table	Key Columns
campaigns	id, name, status, hashtags[], start_date, end_date, type
campaign_posts	id, campaign_id, url, platform, scraped_at, reach, likes, comments, saves, sentiment_score
agencies	id, name, campaign_id, colour_hex
agency_runs	id, campaign_id, run_at, status, apify_run_id
agency_post_scores	id, run_id, agency_id, post_url, reach, likes, comments, saves, perf_score, auth_score, eff_score, total_score, flags[]
fan_pages	id, ig_handle, display_name, followers, last_checked_at, is_active
hashtag_snapshots	id, hashtag, snapshot_at, post_count, avg_eng_rate, top_posts[]

7. Feature Matrix

MVP = minimum viable product launch. Non-MVP features are V2 roadmap items.

Feature	Module	MVP?	Tech Notes
Dashboard KPIs	My Page	Yes	Mock data; later connect IG Graph API
Follower Growth Chart	My Page	Yes	Chart.js line; data from IG API /media
Engagement Breakdown	My Page	Yes	Chart.js doughnut
Content post grid	Content	Yes	IG Media endpoint; thumbnail via CDN URL
Post-type performance	Content	Yes	Aggregate by media_type field
Audience heatmap	Audience	Yes	Best time data; IG Insights endpoint
Location + age charts	Audience	Yes	IG Audience demographics endpoint
Compare Pages	Compare	Yes	Multiple IG accounts via Graph API
Own Campaigns list	Campaigns	Yes	CRUD; store in DB (Firestore/Postgres)
#vijayam detail view	Campaigns	Yes	Hashtag scrape (CrowdTangle/Apify)
Hashtag Search & track	Campaigns	Yes	Hashtag API or scraper; polling job
Agency report upload	Campaigns	Yes	File parse (xlsx/csv) + URL list input
Agency post scraper	Campaigns	No	Apify IG scraper actor; 500-post batch
Authenticity scoring	Campaigns	No	Rule engine; velocity, bot pattern checks
Fan page tracker	Fan Pages	Yes	Manual + auto-detect via hashtag stream
Fan page alerts	Fan Pages	No	Push notification / email on threshold
Real-time hashtag stream	Campaigns	No	WebSocket feed; CrowdTangle or Apify
Sentiment analysis	Campaigns	No	Malayalam NLP model (Google NL API)

8. Open Questions & Decisions Needed

- Instagram Business Account — the Graph API requires the page to be a verified Business or Creator account. Is @nivinpauly already set up? If not, this must be done before any API integration.
- Competitor data access — the Graph API does not expose engagement data for other users' posts. Compare Pages will show only follower counts from public profiles; engagement rates for competitors (Dulquer, etc.) must be estimated or sourced from a third-party service.
- Agency contract terms — does the agency agreement include a clause requiring them to provide post URLs? This is the input for the Agency Report. Without it, the team must manually collect links.
- Who has access — should agencies be able to see their own scores? Or is this internal-only? Determines whether a public-facing report view is needed.
- Alert delivery — for fan page alerts, should the team receive email, WhatsApp, or push notifications? This determines the notification provider (SendGrid, Twilio, Firebase FCM).

Appendix A — Prototype Screen Map

Screen ID	Route (Planned)
dashboard	/
content	/content
audience	/audience
compare	/compare
campaigns/own	/campaigns
campaigns/vijayam	/campaigns/vijayam
campaigns/hashtag	/campaigns/hashtag
campaigns/agency	/campaigns/agency
fan-pages	/fan-pages

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